

Sensitivity of Antarctic sea level contribution to regularisation of the inverse problem used in initialisation

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Abstract. Ice sheet models commonly generate an initial state by solving an inverse problem for basal friction and ice stiffness such that modelled ice velocities closely match observations of the surface. Inverse problems are typically regularised to prevent overfitting to noisy observations, but the appropriate regularisation strength is a subjective modelling choice that is left to the discretion of the user. Even more problematic, inverse problems are ill-posed and can absorb errors originating elsewhere in the model, for example in the bed topography or unresolved physics. While this yields realistic short-term behaviour, it may undermine the accuracy of long-term projections.

Here we assess the sensitivity of sea level contribution from Antarctica to regularisation of the inverse problem used in initialisation.

Here, we assess how regularisation of the inverse problem influences sea level contribution from Antarctica when the model is projected forward. Using the BISICLES ice sheet model, we generate an ensemble of initial states by perturbing two parameters, which regularise the solutions for basal friction and ice stiffness. We then project them forward to 2300 under a high emission scenario. We find that sensitivity to regularisation of the ice-stiffness field is comparable to other commonly perturbed parameters on a century timescale but declines on longer timescales, whereas sensitivity to friction regularisation is consistently small — much lower than reported for an equivalent study of the Amundsen Sea Embayment. Our results suggest that the importance of these initialisation choices for sea level projections may depend on the model, domain, resolution, or timescale considered.

1 Introduction

The Antarctic Ice Sheet is the largest potential contributor to sea level rise, but its response to future climate change remains highly uncertain (Fox-Kemper et al., 2021). This uncertainty is even more pronounced on timescales spanning multiple centuries, with projections to 2300 under high emissions ranging from near-zero to multiple metres of sea level contribution (Golledge et al., 2015; Bulthuis et al., 2019; DeConto et al., 2021; Greve et al., 2023; Seroussi et al., 2024; Coulon et al., 2025). Quantifying uncertainty in the timing and magnitude of sea level rise is important for climate adaptation strategies such as sea defenses, and underestimating uncertainty poses a risk to the resilience of coastal communities (Hinkel et al., 2019;

Kopp et al., 2019). Ice sheet modelling studies that seek to produce usable projections of future sea level rise should therefore
25 make efforts to include the dominant sources of uncertainty.

Uncertainty in the output of ice sheet models arises from numerous sources, including uncertainty in certain model parameters, uncertainty in the climate forcing, and model structural uncertainty, which itself encompasses the form of parameterisations, the stress-balance approximation, the initial state, and much more. Most commonly, studies will consider uncertainty in the climate forcing, either by selecting numerous emission scenarios, climate models, or else through a climate indexing
30 method (e.g. DeConto et al., 2021; Greve et al., 2023). Increasingly, studies are sampling a wide range of uncertain model parameters and parameterisations (Bulthuis et al., 2019; Hill et al., 2021; Coulon et al., 2024, 2025; Rosier et al., 2025). Model structural uncertainty, whilst touched on by some of these studies (notably Coulon et al., 2025), is largely explored through international projects such as the Ice Sheet Model Intercomparison Project (ISMIP6, Nowicki et al., 2016), and is the largest source of uncertainty in those experiments (Seroussi et al., 2023). The InitMIP branch of ISMIP6 was designed to specifically
35 isolate uncertainty in the ice sheet initial state, showing that projected sea level contribution is highly sensitive to the choice of initialisation (Seroussi et al., 2019). However, outside of these large intercomparison projects, uncertainty in the ice sheet initial state is less well-explored, since producing multiple realistic initial states is often time consuming and computationally expensive.

The initial state refers to a set of spatial fields defining the physical condition of the ice sheet, which can include ice thickness, bedrock elevation, ice temperature or enthalpy, basal properties, damage, and others. Ideally, these would be similar to
40 the modern ice sheet whilst producing short-term behaviour that resembles observed trends. However, this is not straightforward to achieve, mainly due to inconsistencies between datasets and the model physics. Initialisation methods that approach this problem can be broadly divided into two philosophies: model spin-ups and data assimilation, although some approaches combine elements of both (Pollard and DeConto, 2012; Le Clec'H et al., 2019). Here, we focus only on initialisation through
45 data assimilation, which accounted for 7 of the 25 submissions to initMIP. Based on early work by MacAyeal (1992), this typically solves an inverse problem for basal friction and ice rheology parameters such that modelled ice velocities closely match observations. Approaches of this kind have been implemented into numerous ice sheet models and applied to the Antarctic Ice Sheet or its basins (Morlighem et al., 2013; Gagliardini et al., 2013; Cornford et al., 2015; Arthern et al., 2015; Hoffman et al., 2018; Rosier et al., 2025). Inverse methods can produce initial states with very good fits to observed dynamics. However,
50 they are also ill-posed in that the data does not uniquely constrain the solution. Other equally plausible solutions to the inverse problem could yield different future sea level outcomes, but these are rarely explored. Furthermore, inverse problems can introduce compensating errors for deficiencies elsewhere in the model, which then influence future mass loss when the model is projected forward (Berends et al., 2023).

Standard approaches to the inverse problem minimise a loss function comprised of a model-observation misfit and regularisation term. This yields a single best estimate but gives no information about its associated uncertainty. For this, one would
55 need to know the local curvature of the loss function about its minimum, which is described by the Hessian matrix. For practical ice sheet modelling purposes, the Hessian is too large to compute directly, but can be approximated via its low-rank structure (Petra et al., 2014). The resulting problem is still expensive to solve, but has allowed for the local approximation of

the posterior distribution in idealised ice sheet problems (Petra et al., 2014; Babaniyi et al., 2021) or small realistic domains (Gopalan et al., 2021; Brinkerhoff, 2022). Isaac et al. (2015) perform a Bayesian inversion for basal friction over the Antarctic ice sheet by further assuming that the posterior is locally Gaussian about the optimal solution (equivalently: the loss function is locally quadratic). However, this may not translate well to inverse problems that solve for both friction and rheology, since non-uniqueness of the solution could produce posterior geometry that is poorly represented by a Gaussian approximation. A Bayesian inversion for both friction and rheology has yet to be completed at a continent scale, motivating alternative methods that do not sample from the posterior, but nonetheless explore uncertainty in the solution to the inverse problem. The simplest approach is to produce an ensemble of projections in which the solved fields from a single inversion are variably scaled by a uniform multiplier (e.g. Nias et al., 2016; Bevan et al., 2023). These studies then downweight ensemble members with poor fits to observed dynamics using a post-hoc calibration. However, this only allows for uniform scaling and leaves the spatial pattern unchanged. Some studies perturb the bedrock elevation and solve inverse problems for each realisation (Werneck et al., 2022; Krishna et al., 2026), though care must be taken to account for spatially autocorrelated errors in the topography dataset. A third approach is to treat hyperparameters within the inversion as uncertain parameters and perturb these in a traditional uncertainty quantification framework (Rosier et al., 2025). This allows for spatially-varying perturbations that are physically plausible by design.

Methods to solve the inverse problem vary between models (see e.g. Barnes et al., 2021), but most rely on some form of regularisation. If unregularised, the optimisation introduces short-wavelength, high-amplitude variability into the inferred fields, which overfit to noisy errors in the observed velocity. Regularisation imposes structure on the solution by penalising roughness and encouraging smooth solutions. However, the strength of regularisation is left largely up to the discretion of the user. An L-curve heuristic is commonly used to justify a choice of regularisation strength by finding the optimal compromise between solution fit and smoothness, characterised by the corner of the ‘L’ (Hansen, 1992). This does not guarantee, however, that the corner of the L-curve corresponds to anything physically meaningful, nor that the resulting fields are representative of real basal and rheological properties. The location of the L-curve corner has also been shown to depend on choices such as mesh resolution and parameterisations, which complicates its use as an objective criterion (Wolovick et al., 2023). Furthermore, Rosier et al. (2025) demonstrate that projected sea level contribution from the Amundsen Sea Sector can be highly sensitive to regularisation in the inverse problem as formalised by the Úa ice sheet model. This suggests that regularisation hyperparameters may be a route to exploring uncertainty in sea level projections arising from the inverse problem — and, by association, the initial state — but it is not yet clear how this result translates to other models or domains.

To further study uncertainties associated with the inverse problem, we use the BISICLES ice sheet model to assess the sensitivity of Antarctic sea level contribution to regularisation parameters in the inverse problem. We produce an ensemble of initial states by perturbing these parameters, alongside other commonly perturbed ice sheet model parameters, and project them forward to 2300 under a high-emission scenario. We find that regularisation parameters can be a moderate source of uncertainty in century-scale sea level projections, but that this sensitivity is likely not consistent across models, domains, and timescales. We discuss the implications of these results for future research efforts to explore initial state uncertainty in sea level projections.

2 Methods

95 2.1 BISICLES ice sheet model

BISICLES (Cornford et al., 2013) is a finite volume, vertically integrated ice sheet model, which has been used extensively for century-scale modelling of the Antarctic Ice Sheet (Cornford et al., 2015; Nias et al., 2019; Sun et al., 2020; Wernecke et al., 2022; Siahhaan et al., 2022; O’Neill et al., 2025). It uses the LIL2 approximation to Stokes flow (Schoof and Hindmarsh, 2010), which incorporates membrane stresses with a simplified treatment of vertical shear, allowing it to simulate a wide range of flow regimes in Antarctica with computational efficiency. It also features an adaptive mesh, which refines mesh resolution close to the grounding line and in regions of fast flow. Here, we run BISICLES with a base resolution of 8 km and 3 levels of refinement, up to a finest resolution of 1 km. This significantly improves the representation of grounding line dynamics compared to coarser grids (Cornford et al., 2016).

BISICLES solves the 2-dimensional LIL2 mass transport and stress balance equations outlined in Cornford et al. (2015). They feature a vertically integrated effective viscosity, $\phi h \bar{\mu}$, defined by:

$$\phi h \bar{\mu}(x, y) = \phi \int_{s-h}^s \mu(x, y, z) dz, \quad (1)$$

where h is ice thickness, s is surface elevation, and ϕ is an uncertain, spatially-varying coefficient that is determined by the inverse problem. Values of $\phi < 1$ correspond to softening of the ice and values $\phi > 1$ to strengthening, with a default value $\phi = 1$. The vertically varying effective viscosity, $\mu(x, y, z)$, is calculated using Glen’s flow law (Glen, 1955):

$$110 \quad \mu(x, y, z) = \frac{1}{2A_0 A(T) \tau^{n-1}}, \quad (2)$$

where $A(T)$ is a temperature-dependent Paterson rate factor (Cuffey and Paterson, 2010), τ is the effective deviatoric stress, and the flow law exponent n is an uncertain parameter that we perturb in our experiment. The 3-dimensional temperature field is derived from a 100,000-year thermal spin-up described in Trevers et al. (2024). Previous studies have perturbed n after model initialisation by scaling $A(x, y)$ to ensure that the effective viscosity is unchanged everywhere (Getraer and Morlighem, 2025; Rosier et al., 2025). However, we follow (Martin et al., 2026) by fixing n before initialisation and introducing a spatially-uniform correction factor, $A_0 = 100,000^{(3-n)}$ to ensure that the effective viscosity is independent of n at a reference stress of 100 kPa. Either side of this value, the flow law diverges from the $n = 3$ case, but the effective viscosity is scaled by ϕ during the inverse problem so that the initial state matches observed ice velocities.

Basal sliding is governed by a regularised-Coulomb sliding law, which smoothly transitions between Weertman and Coulomb behaviour near to a reference velocity $u_0 = 300 \text{ m a}^{-1}$ (Joughin et al., 2019). As such, the slow-moving ice sheet interior stays in a power-law regime, whilst fast-flowing ice streams approach the perfectly plastic sliding limit. The basal shear stress, $\tau_b = -\alpha \mathbf{u}_b$, is given by:

$$\alpha = C |\mathbf{u}_b|^{m-1} \left(\frac{|\mathbf{u}_b|}{u_0} + 1 \right)^{-m}, \quad (3)$$

where the friction exponent m is an uncertain parameter that we perturb in our experiment and C is a spatially-varying friction coefficient that is also determined by the inverse problem. The form in (3) is equivalent to Joughin et al. (2019) with $m \rightarrow 1/m$ and $C \rightarrow Cu_0^m$, giving C the same units as in the standard power law. This performs similarly to other commonly used sliding laws for key glaciers in Antarctica (Barnes and Gudmundsson, 2022).

2.2 Model initialisation

We initialise BISICLES using an inverse method to solve for the basal friction coefficient, C , and the viscosity coefficient, ϕ , such that modelled surface velocities closely match observations. BISICLES uses conjugate gradient descent (Shewchuk, 1994) to minimise a cost function, J , defined by:

$$J = J_m + J_p, \quad (4)$$

where J_m is a function of the velocity misfit

$$J_m = \frac{1}{2} \int_{\Omega_V} (|\mathbf{u}| - |\mathbf{u}_{obs}|)^2 d\Omega_V, \quad (5)$$

and J_p is a Tikhonov penalty function

$$J_p = \frac{\alpha_C}{2} \int_{\Omega_V} |\nabla C|^2 d\Omega_V + \frac{\alpha_\phi}{2} \int_{\Omega_V} |\nabla \phi|^2 d\Omega_V, \quad (6)$$

where α_C and α_ϕ are regularisation parameters that respectively control the smoothness of C and ϕ , Ω_V is the domain over which velocities are observed, and $\mathbf{u}_{obs}$ are the surface velocities from Rignot et al. (2017). In our experiment, we treat both α_C and α_ϕ as uncertain parameters within a range of values (Section 2.4).

We solve the inverse problem using a linear sliding law ($\tau_b = -C_1 \mathbf{u}_b$) to simplify the stress balance. C is then adjusted post-inversion such that basal drag (and therefore ice velocity) remains unchanged. This is done by setting

$$C_m = C_1 |\mathbf{u}|^{1-m} \left(\frac{|\mathbf{u}|}{u_0} + 1 \right)^m. \quad (7)$$

Here, a minimum of 1 ma^{-1} is applied over $|\mathbf{u}(x, y)|$ to avoid unrealistically high friction in regions of very slow flow. This has the drawback that the initial velocity of ice in regions with $|\mathbf{u}| \ll 1$ may no longer be an optimal fit to observations, but only occurs in regions in which the observed velocity already poorly constrains basal friction (Isaac et al., 2015). The initial ice geometry is taken from BedMachine Antarctica v3 (Morlighem, 2022), the initial guess for ϕ is uniformly 1, and the initial guess for C is calculated by equating basal drag to the driving stress. This yields

$$C_{init} = \rho_i g h |\nabla s| |\mathbf{u}|^{-1}, \quad (8)$$

where $\rho_i = 917 \text{ kg m}^{-3}$ is the density of ice and $g = 9.81 \text{ m s}^{-2}$ is gravitational acceleration. As is common with initialisation by inverse problem, we observe a short-term shock immediately after initialisation as the model adjusts to inconsistencies

between the geometry and model physics. We therefore allow a 15-year relaxation period, during which we periodically re-solve the inverse problem each year following Bevan et al. (2023). Throughout the relaxation, the thickness of floating ice shelves is held constant and we apply a constant background surface mass balance, derived by taking the 1980–2021 mean surface mass balance from a simulation by the MAR regional climate model, forced by ERA5-Interim data (Agosta et al., 2019). **NOTE: I got this dataset from Matt — Matt, do you know if this is the correct citation and have I described this correctly? Because Agosta et al. only seems to go up to 2015.** In total, we generate 7 initial states for our experiment by perturbing parameters used within the initialisation (see Section 2.4). These are then projected forward under a climate forcing described in Section 2.3.

2.3 Climate forcing

Following initialisation, we apply a time-varying surface and basal mass balance between an initial date in 2007 until 2300. The surface mass balance is composed of the background rate used in the relaxation, upon which we apply anomalies derived from the CESM2-WACCM climate model (Danabasoglu et al., 2020) under the SSP5-8.5 emissions scenario (Meinshausen et al., 2020). The anomalies are annual means with respect to the 1995–2014 climatology, calculated by subtracting evaporation, sublimation and runoff from precipitation, as in (Nowicki et al., 2020). CESM2-WACCM was chosen for its strong climate forcing to maximise the signal-to-noise ratio in our results, but its high climate sensitivity and the aggressive increase in emissions in SSP5-8.5 mean it is not necessarily representative of the most likely future climate (Hausfather, 2025). **TODO: ref for CESM2 climate sensitivity**

To compute a basal mass balance, we follow the method described by Jourdain et al. (2020), using the non-local quadratic parameterisation of ice shelf basal melt with respect to ocean thermal forcing.

$$m(x, y) = \gamma_0 \times \left(\frac{\rho_{\text{sw}} c_{\text{pw}}}{\rho_i L_f} \right)^2 \quad (9)$$

$$\times (\text{TF}(x, y, z_{\text{draft}}) + \delta T_{\text{sector}}) \quad (10)$$

$$\times |\langle \text{TF} \rangle_{\text{draft} \in \text{sector}} + \delta T_{\text{sector}}|, \quad (11)$$

where ρ_{sw} and c_{pw} are the density and specific heat capacity of seawater, ρ_i is the density of ice, L_f is the latent heat of fusion, and γ_0 is the ice-ocean heat flux. $\text{TF}(x, y, z_{\text{draft}})$ is the ocean thermal forcing anomaly at the depth of the ice shelf draft, calculated from ocean temperature and salinity outputs from CESM2-WACCM. Anomalies are calculated with respect to the initial thermal forcing such that basal mass balance is zero upon initialisation. Whilst this underestimates present-day melt rates in some key regions (Adusumilli et al., 2020), any errors are quickly outweighed by the strong climate forcing in SSP5-8.5 and we consider it to be adequate for the purposes of a sensitivity study. δT_{sector} is a sector-specific offset used by Jourdain et al. (2020) to fit basal melt to observed rates. However, our anomaly approach significantly reduces the impact of this term and we fix it constant at the ‘MeanAnt’ median-calibrated values derived by Jourdain et al. (2020).

Table 1. Perturbed parameters used in the ensemble.

Parameter	Long name	Min.	Nominal	Max.
α_C	C -regularisation	10^{-2}	10^0	10^2
α_ϕ	ϕ -regularisation	10^8	10^{10}	10^{12}
n	Flow law exponent	2	3	4
m	Sliding exponent	1	3	5
γ_0	Ice-ocean heat flux	9619	39131	159189

2.4 Experiment design

We assess sensitivity to parameters through a simple ‘axial’ ensemble, in which we perturb one parameter at a time to a given upper or lower bound whilst holding all other parameters at their chosen nominal value (Table 1). A method suggested by Wolovick et al. (2023) chooses a suitable range of regularisation values by considering the points at which L-curve curvature has dropped to a half of its maximum value (the maximum occurring at the L-curve corner). However, the authors note that the choice of a half is arbitrary and that the upper bound in particular is highly sensitive to this choice. In the interest of sampling the widest possible uncertainty, we instead choose an upper and lower bound based on separate criteria: the lower bound is selected such that lower values of regularisation no longer improve the misfit (the solution is essentially unregularised); the upper bound corresponds to a solution which offers no (or marginal) improvement to the misfit over the initial guess. Note that, for C , this does not mean that the solution is equal to the driving stress, since smoother solutions can still improve the misfit. However, for ϕ , this results in a solution that is near-uniformly 1. This level of regularisation is well beyond the level suggested by L-curve heuristics, but represents a case in which the model’s inability to capture processes normally absorbed by ϕ (macroscopic damage, crystal anisotropy, etc.) is laid bare.

For comparison with the regularisation parameters, we consider three other commonly perturbed ice sheet model parameters: the flow law exponent, n , the sliding exponent, m , and the ice-ocean heat flux, γ_0 . The nominal value of each parameter corresponds to either the arithmetic (m , n) or geometric (γ_0) mean of its upper and lower bounds. These are chosen to be representative of typical values used across a range of studies (e.g. Nias et al., 2019; Barnes and Gudmundsson, 2022; Wolovick et al., 2023; Seroussi et al., 2024; Coulon et al., 2024, 2025; Rosier et al., 2025; Getraer and Morlighem, 2025). All in all, we perturb five parameters in BISICLES across a range of different model components. This gives us a total of 10 forward simulations from 6 initial states (some initial states are shared between ensemble members). This is a much simpler design than other space-filling options, which also allow for more sophisticated sensitivity analysis such as Sobol’ sensitivity (Hill et al., 2021; Rosier et al., 2025). However, it is sufficient for a preliminary exploration of sensitivity before investing in a more comprehensive experiment design.

3 Results

205 The impact of high and low regularisation on the solutions for C and ϕ , as well as the corresponding velocity misfits, are shown in Figure 1. Minimal regularisation produces a solution for C with high spatial variability (Figure 1a), taking very high values ($> 10^4$) in regions of slow flow and very low values ($< 10^1$) in fast-flowing ice streams. This results in a good fit to observed velocities, where the regions of highest misfit are over the Trans-Antarctic mountains and the Antarctic Peninsula. As one would expect, high regularisation produces a comparatively smoother solution for C (Figure 1b) with a worse fit to
 210 observed velocities. The high values of C in slow-flowing regions are approximately halved and the low values of C in fast-flowing regions are approximately doubled. As a result, the model slightly underestimates ice velocity in the ice streams and overestimates ice velocity over much of the rest of the continent.

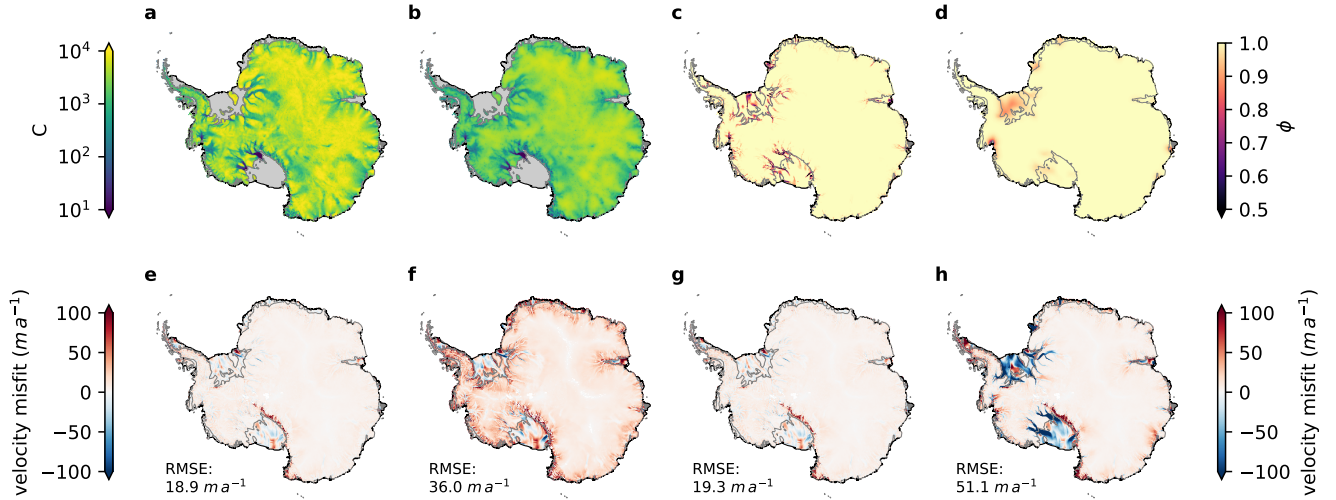


Figure 1. a–d: Solutions to the inverse problem for ensemble members with low (a, c) and high (b, d) regularisation of C (a, b) and ϕ (c, d). e–h: The corresponding velocity misfit for each solution, both spatially and aggregated using RMSE. The grounding line and ice shelf fronts are outlined in black.

The minimally-regularised solution for ϕ (Figure 1c) stays close to its initial guess of $\phi = 1$ in most regions except for in ice stream shear margins, the grounding zone, and some ice shelves. In these regions, ϕ has very fine spatial structure, taking
 215 values between 0.5 and 1 (i.e. halving the effective viscosity in some areas). This corresponds well with the expectation that ϕ should account for ice damage, which we broadly expect to occur in regions of high strain. However, we make no effort to validate this against remote sensing observations of ice damage and note that similar approaches have found inverse methods to also predict ice damage in regions where it is not observed (Gerli et al., 2024) **NOTE: Is it better to not mention this in case we're asked to do some validation?** When ϕ is highly regularised (Figure 1d), this spatial structure is smoothed out and
 220 ϕ only takes values visibly less than 1 in the vicinity of Pine Island Glacier and broadly over the Filchner-Ronne ice shelf. This

significantly increases the velocity misfit, particularly in the ice streams feeding the Filchner-Ronne and Ross ice shelves, in which velocity is underestimated by over 100 m a^{-1} .

The initial states that correspond with these four solutions, as well as initial states for our other perturbed parameters, are projected forward under the climate forcing described in Section 2.3. The resulting sea level rise contributed by each ensemble member is shown in Figure 2. The ensemble range at 2100 is 1.1–22.4 cm, which lies in the middle of the ISMIP6 range of -7.8–30.0 cm (Seroussi et al., 2020). In the 22nd and 23rd centuries, all ensemble members accelerate their mass loss and the range widens to 0.84–1.97 m by 2300. Our range then sits in the lower half of the ISMIP-style multi-model ensemble by Seroussi et al. (2024), although we note that, by perturbing only one parameter at a time from their nominal value, we are likely missing some parameter combinations that could produce substantially greater sea level contribution from Antarctica.

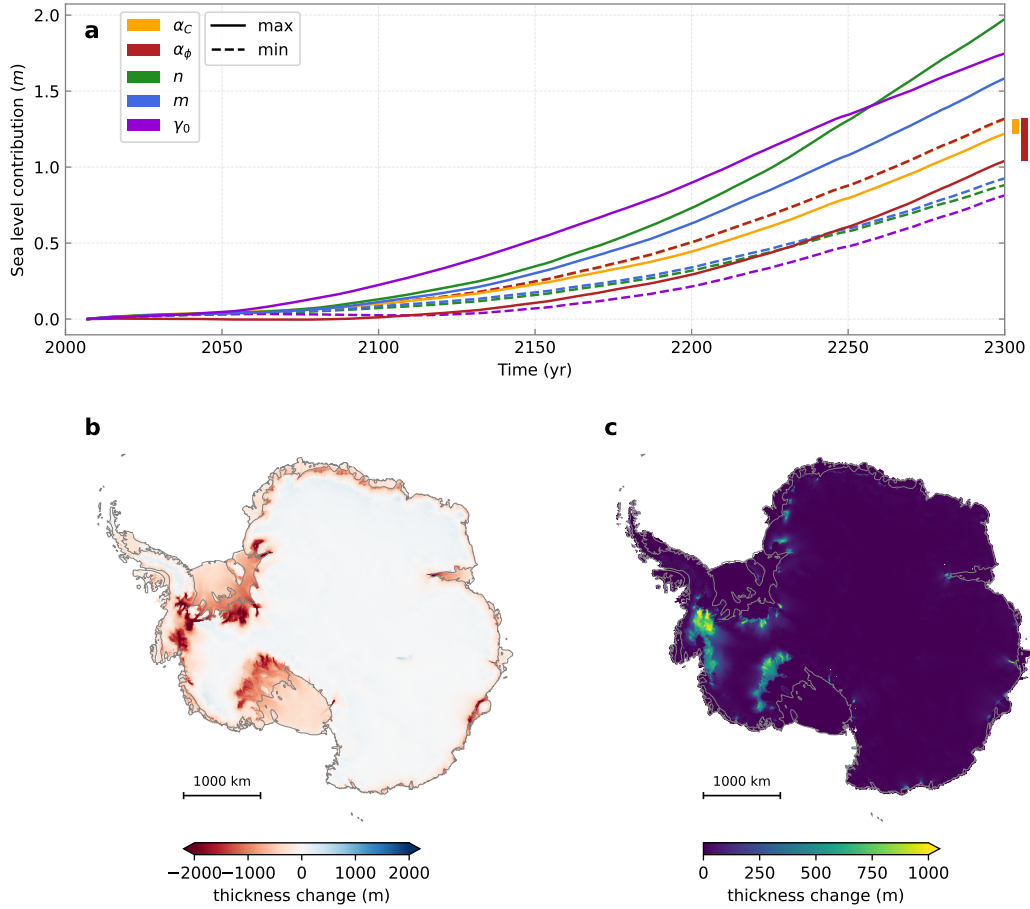


Figure 2. (a): Sea level rise contributed by each ensemble member between 2007–2300. Ensemble members that correspond to a perturbed parameter are labelled by their colour. Solid lines correspond to a parameter at its maximal value; dashed lines to a parameter at its minimal value. The range in SLC by 2300 is shown by coloured bars (right). Below: the ensemble mean (b) and standard deviation (c) of ice thickness change in Antarctica between 2007 and 2300. The grounding line and ice shelf fronts are outlined in grey.

230 For each parameter, we calculate the range in sea level contribution at 2300 between ensemble members with minimal and maximal values of that parameter. This provides a rudimentary measure of sensitivity, which we can compare across parameters (coloured bars in Figure 2a). The largest range in sea level contribution is produced by perturbing the flow law exponent n , followed by the ice-ocean heat flux γ_0 and friction exponent m . The regularisation parameters, α_ϕ and α_C , comparatively produce much smaller ranges **TODO: put in actual values here**. Higher regularisation of the inverse problem (both C and ϕ)
 235 leads to lower sea level contribution, consistent with the previous observation that very high regularisation produces slower ice streams and therefore lower grounding line flux.

The spatial patterns of ice thickness change are broadly similar across the ensemble, with most mass loss occurring from West Antarctica (Figure 2b). As also found by Seroussi et al. (2024), the major ice shelves typically collapse during the 22nd century, although the timing of collapse varies by up to 40 years depending on our value of γ_0 . The Amundsen Sea Embayment
 240 and the Siple Coast are two major regions of mass loss, as are numerous tributaries of the Filchner and Ronne Ice Shelves. Pine Island Glacier in particular retreats on average ~ 250 km upstream of its present-day grounding line. In comparison, Thwaites Glacier thinning is more limited and does not propagate far upstream. Some of these regions are sensitive to parameter choices, indicated by high variance in ice thickness change across the ensemble (Figure 2c). The Siple Coast retreats in all ensemble members, but the extent of retreat varies by 100–200 km. Similarly, the extent of retreat into the Pine Island and Rutford Glacier
 245 basins vary strongly with parameter choices. **TODO: Thwaites variance**.

Parameter sensitivities similar to those above can be calculated at the end of each century, allowing us to assess how sensitivities change over time (Figure 3). At 2100, the largest sensitivity is to the ice-ocean heat flux, γ_0 . However, after the major ice shelves collapse in the 22nd century, the role of γ_0 diminishes and other parameters become more important. The parameters involved in the stress balance, n and m , are relatively unimportant at 2100, but their sensitivities increase over time with larger
 250 changes in ice geometry driven by ocean melt. Compared to these parameters, we find that sea level contribution is relatively insensitive to regularisation of basal friction (α_C) across all timescales. However, it is highly sensitive to regularisation of the viscosity coefficient (α_ϕ) at 2100, second only to γ_0 . This sensitivity, however, decreases over time as the ice sheet retreats into regions where ϕ is mostly uniform. By 2300, the sensitivity to both regularisation parameters falls behind n , m , and γ_0 .

4 Discussion

255 In this study, we have assessed the sensitivity of Antarctic sea level projections to regularisation parameters in the inverse problem used to initialise the BISICLES ice sheet model. In our experiment, we find that regularisation of the basal friction coefficient, C , is a relatively small source of uncertainty in multi-century sea level projections. We also find that regularisation of the viscosity coefficient, ϕ , is a notable source of uncertainty on century timescales, but that its contribution to uncertainty decreases over time. Our results compare directly with those of Rosier et al. (2025), who also perturb regularisation param-
 260 eters in experiments with the Úa ice sheet model applied to the Amundsen Sea Embayment. Rosier et al. (2025) find that regularisation parameters related to both basal friction and rheology are significant sources of uncertainty in 21st century sea level projections. Our results provide some context to these findings, suggesting that the sensitivity of sea level contribution

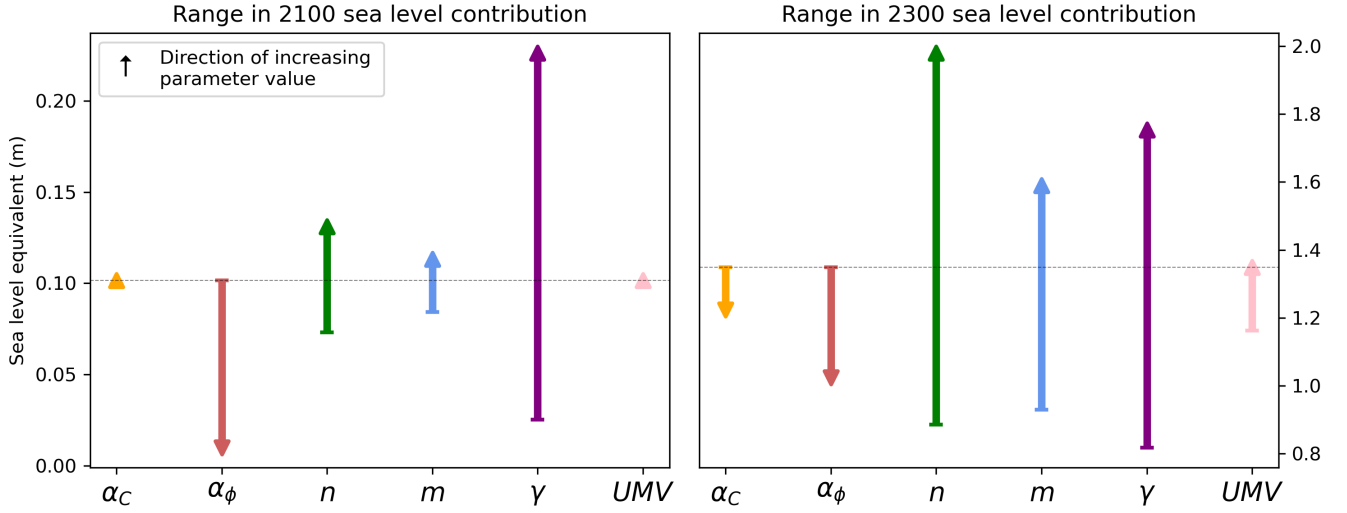


Figure 3. Sensitivity of Antarctic sea level contribution to each perturbed parameter at (a) 2100 and (b) 2300. Arrows denote the range in sea level contribution between ensemble members with minimal and maximal values of that parameter, pointing from minimal to maximal. Arrows pointing upwards/downwards therefore indicate a parameter that increases/decreases sea level contribution with higher values. The dashed horizontal line indicates the sea level contribution in a simulation with all parameters held at their default value.

to regularisation parameters may decrease over longer timescales. We will first address the differences between our results with regard to regularisation of rheology parameters, before discussing the differences in sensitivity to regularisation of basal friction.

As we have discussed, Ice sheet models commonly invert for a parameter relating to rheology as well as a basal friction coefficient. In our case, that parameter is the viscosity coefficient, ϕ , though $\dot{U}a$ simply inverts directly for the rate factor, A . These parameters are typically important in the shear margins, ice shelves, and grounding zone, where high shear or extensional stresses can induce high rates of damage. In reality, the damage processes that these parameters absorb (among others) are likely to propagate inland as the grounding line retreats. However, this is not captured by the inverse problem, which provides only a static solution based on modern ice geometry. As a result, the sensitivity to regularisation of these parameters decreases once the ice sheet has retreated into regions that are currently undamaged ($\phi \approx 1$). This framing also highlights the importance of developing ice sheet models that can capture damage processes. We show that severely limiting ϕ results in much slower ice velocities and lower sea level contribution. For the same reason, the lack of a coupled damage model in BISICLES and other ice sheet models may underestimate sea level projections when the ice sheet retreats deep into regions past the present-day grounding line. **NOTE: Have I leaned too much into damage here? Maybe I should just talk about the role of ϕ in “different stress regimes”, which is perhaps a more broad framing.**

Compared to $\dot{U}a$, BISICLES is much less sensitive to regularisation of basal friction. The most likely explanation for this is the different forms of regularisation used between the two models. BISICLES regularises by penalising high values of ∇C ,

280 a form which is also used by several other ice sheet models (Morlighem et al., 2013; Gagliardini et al., 2013; Hoffman et al., 2018). The equivalent term in $\dot{U}a$ takes the form $\nabla \log(C/\tilde{C})$, where $\tilde{C}$ is the initial guess. Since C varies over several orders of magnitude, this means that the penalty in BISICLES is dominated by regions of high friction, which are typically slow-moving and therefore less dynamically important to century-scale sea level projections. In $\dot{U}a$, the friction is both normalised by the initial guess and log-transformed, which reduces the influence of high friction values and focuses the penalty on regions of low friction such as the ice streams. The difference in model domains may also play a role, with our inclusion of large slow-moving regions of East Antarctica likely reducing sensitivity to friction regularisation relative to the Amundsen Sea Embayment.

Our results can also be contextualised by the broader literature assessing the sensitivity of the Antarctic Ice Sheet to numerous ice sheet model parameters. Very few of these have considered uncertainty in the initial state, and only Rosier et al. (2025) have done this by perturbing regularisation parameters in the inverse problem. However, the other parameters perturbed in our ensemble have a much higher representation in the literature and can be loosely compared with our results, albeit over various different domains and timescales. For example, Barnes and Gudmundsson (2022) and Getraer and Morlighem (2025) perturb m and n respectively in simulations of the Amundsen Sea Embayment. The sensitivity of sea level contribution as m is increased from 1 to 5 and n is increased from 3 to 4 is broadly consistent with our results for the whole of Antarctica. Coulon et al. (2025) use the same quadratic non-local parameterisation of basal melt (as well as several alternative parameterisations), but perturb γ_0 over a narrower range. Their models produce much greater sea level contribution by 2300 than ours, but this is largely due to the much wider sampling of uncertainty in other aspects of the model. A significant portion of sea level contribution in their ensemble comes from Thwaites Glacier, which is much more stable in our simulations. Previous experiments with BISICLES have shown that Thwaites Glacier is very sensitive to the choice of grid resolution, often requiring a resolution of 500 m (i.e. finer than our 1 km mesh) to initiate collapse (Cornford et al., 2016). If Thwaites Glacier were to collapse in our simulations, it would likely have a significant impact on the sensitivity of all parameters.

As with all measures of sensitivity, our results are highly dependent on the range of values within which we choose to perturb each parameter. For this reason, we deliberately choose regularisation values that are well beyond the range supported by L-curve heuristics (Wolovick et al., 2023). In the case of extremely high regularisation, this yields initial velocities that are clearly far from observed values and may be considered unacceptable, particularly for short-term projections. However, we see that, even with this extreme range of values, the impacts of these regularisation parameters on multi-century sea level projections are still much lower than other parameters that are already commonly perturbed in ice sheet model ensembles. We therefore conclude that our results are robust to the choice of parameter ranges and that regularisation of the inverse problem (as formalised above) is a relatively small source of uncertainty in multi-century Antarctic sea level projections compared to other commonly perturbed parameters. However, as a proxy for initial state uncertainty, this is still a very narrow exploration of all possible initial states. Excluded from our experiment are other methods of initialisation, such as palaeoclimate spin-ups, which tend to produce initial states with much more widely varying ice geometry and therefore a wide range of future trajectories (Seroussi et al., 2019).

5 Conclusions

We have explored the sensitivity of Antarctic sea level contribution to regularisation hyperparameters in the inverse problem and compared it with other commonly perturbed parameters. Using the BISICLES ice sheet model, the inverse problem solves for a spatially-varying friction coefficient (C) and viscosity coefficient (ϕ). We find that sensitivity to friction-regularisation is much lower than in an equivalent study for the Amundsen Sea Embayment and suggest that this is due to a difference in model domains and resolution, particularly the inclusion of large slow-moving regions of East Antarctica in our experiment. We find that sensitivity to viscosity-regularisation is high relative to other parameters on a century-timescale, but that this sensitivity declines over time as dynamical parameters dominate the multi-century response. This remains a valid approach to exploring initial state uncertainty in modelling experiments that initialise using inverse methods. However, we advise future experiments of this kind to consider the model, domain, resolution, and timescale of their experiment before choosing parameters to perturb, and warn that even if these parameters are included, it may not even then fully sample the uncertainty associated with the inverse problem.

325 *Code availability.* TEXT

Data availability. TEXT

Code and data availability. TEXT

Sample availability. TEXT

Video supplement. TEXT

330 *Author contributions.* TEXT

Competing interests. TEXT

Disclaimer. TEXT

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